20x20 Sensing Points Seat Cushion Pressure Sensor Specification

Solution Overview----------2

Sensor Structure Description----------3

Performance Parameters----------4

Working Principle and Characteristics---------- ----------5

Pressure Sensing Image Processing Algorithm----------6

How To Connect The Device Properly?----------7

How To Start The Software Properly?----------8

Graphic Settings----------9

Data Storage Interface----------10

Different Mapping Style----------11

Scenarios of failure---------- -12,13

Frequently Asked Questions----------14,15

Solution Overview

• Product Definition and Core Attributes

Maxrays fabric pressure sensor is a flexible pressure-sensing device with fabric as its core carrier, integrating dual characteristics:

Fabric properties: It possesses the flexibility, breathability, and wearability of traditional fabrics, making it suitable for human contact or flexible scenarios;

Sensing function: It can sense pressure changes and convert them into electrical signals, enabling quantitative collection of pressure information.



• Application Scenarios

With its flexibility, wearability, and pressure-sensing properties, it is widely applicable in multiple fields:

Health physiotherapy and medical equipment (e.g., human body pressure monitoring, rehabilitation devices);

Smart home and consumer electronics (e.g., smart mattresses, wearable devices);

Automotive electronics (e.g., seat pressure sensing, safety monitoring);

Industrial control (e.g., pressure detection in automated equipment), etc.

• Usage Advantages

Materials and craftsmanship: Made of flexible materials, it can be attached to the surface or inside of curved products through a thermal lamination process, adapting to measured objects with complex shapes

Ease of use: Pressure readings can be obtained simply by placing it on the surface of the measured part, making operation convenient.

All layers are in full compliance with RoHS environmental testing requirements., No odorless and free ofharmful substances such as halogens, heavy metals, phenols, and aldehydes

Sensor Structure Description

- The main body of the sensor is divided into five layers, with the specific structure as follows:

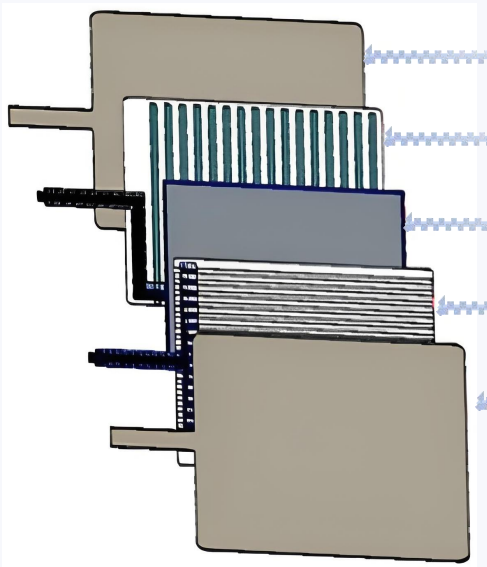
Layer 1: Exterior fabric

Layer 2: Y electrode layer (composed of white fabric, conductive fabric, and PET film with screen-printed circuits)

Layer 3: Piezoresistive film

Layer 4: X electrode layer (composed of white fabric, conductive fabric, and PET film with screen-printed circuits)

Layer 5: Exterior fabric

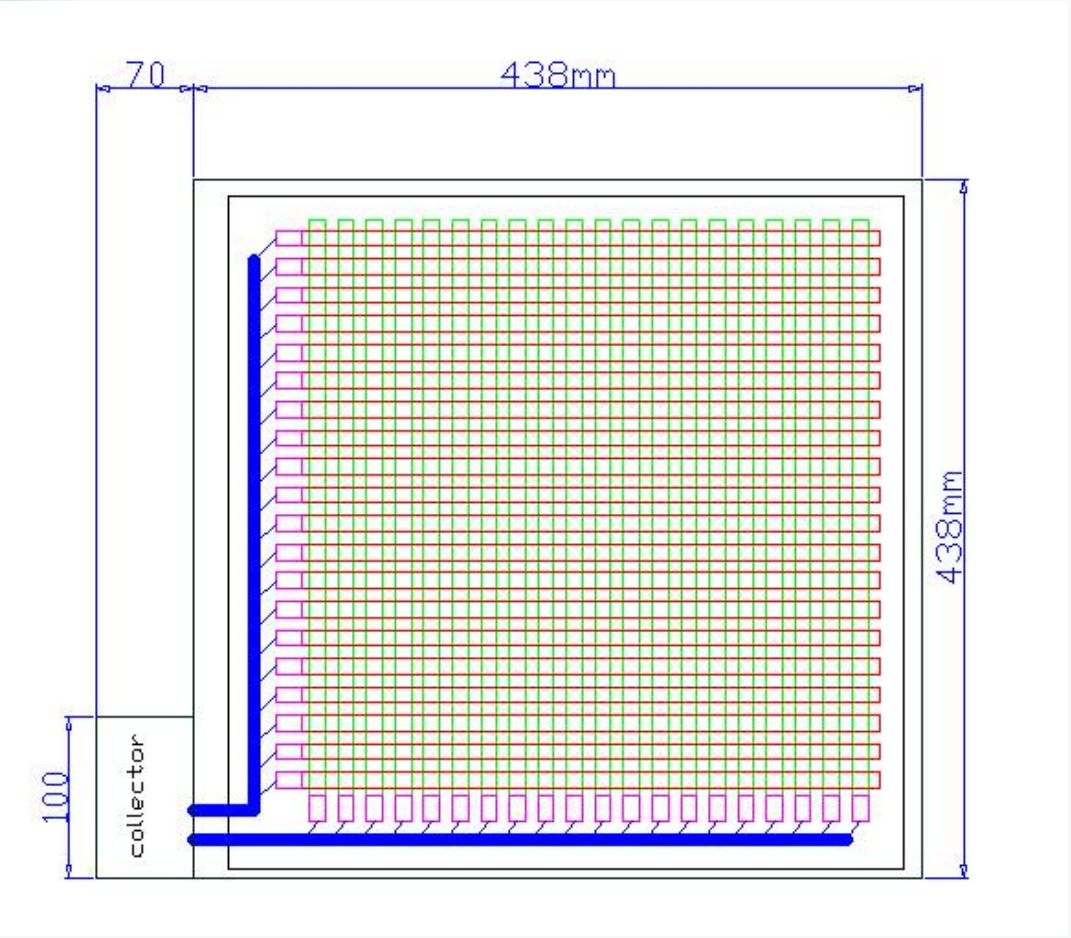


Materials between layers can be assembled by two methods: water-based adhesive bonding and sewing.

- Customization Instruction:

The appearance of the sensor can be customized according to customer needs. The texture and color of the fabric can be provided by the customer or customized on demand.

Performance Parameters



ITEM	Performance parameters
Appearance Dimensions	See the structural diagram
thickness	＜5mm
Static resistance	>0.2MΩ
Range	800Pa-200KPa (1 cm2 circular contact measurement)
Repeatability	+/-15% (1.5Kg load)
Product consistency	+/-20%（1.5Kg load）
Operating temperature	-20° C～60° C
Operating humidity	0～95%RH
Operating voltage	5V
Maximum flow	200mA
Warranty Period	1 year
Communication method	Type C to USB

• Working Principle and Characteristics

Pressure-resistance conversion mechanism:

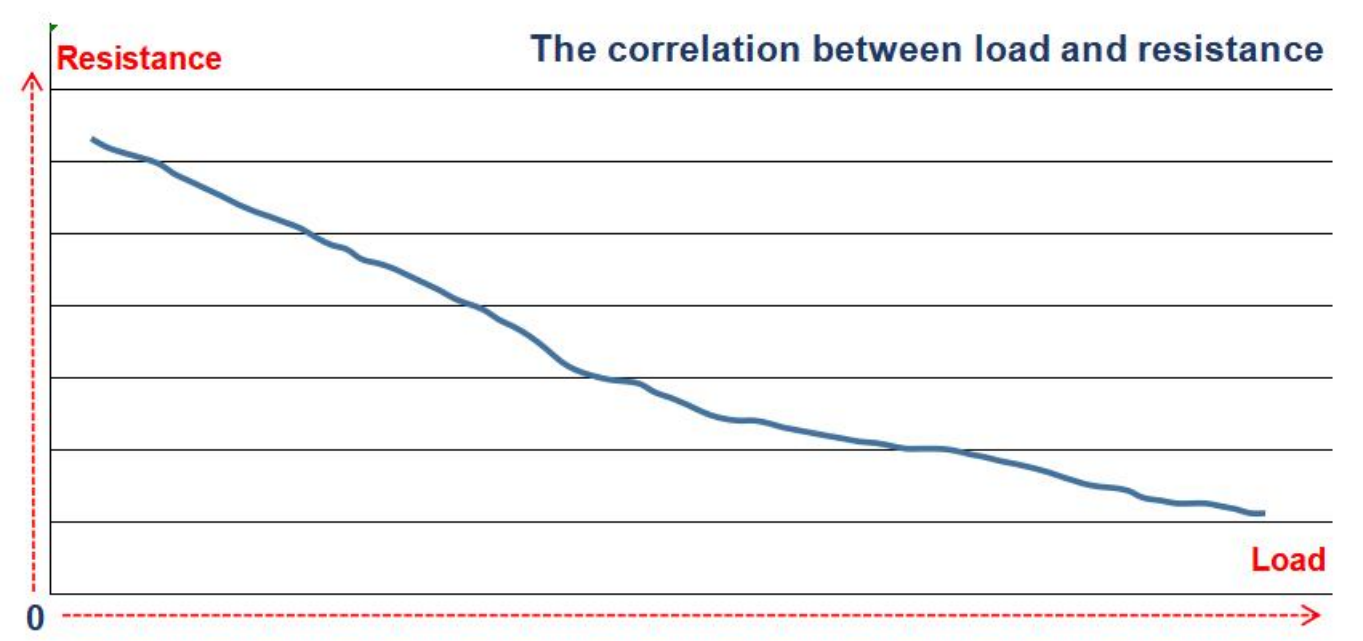
The pressure applied to the sensor area is converted into a change in resistance value (the greater the pressure, the lower the resistance), thereby realizing the acquisition of pressure information.

Mathematical relationships:

Resistance and pressure intensity show a power function relationship;

The reciprocal of resistance (i.e., conductance) has an approximately linear relationship with pressure, a characteristic that helps simplify the processing and analysis of pressure data.

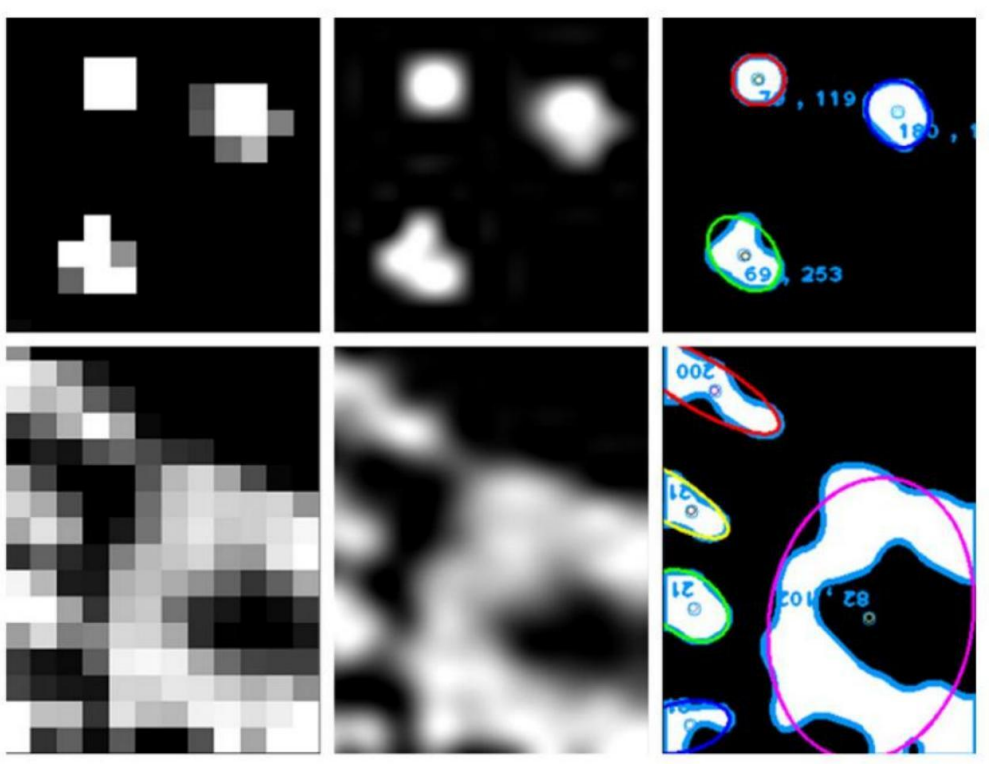
Pressure-resistance correlation curve (for reference only)



The test uses a circular indenter with a diameter of 10mm as the acting area, and the buffer pad is silicone with a thickness of 0.9mm.

- **Pressure Sensing Image Processing Algorithm**

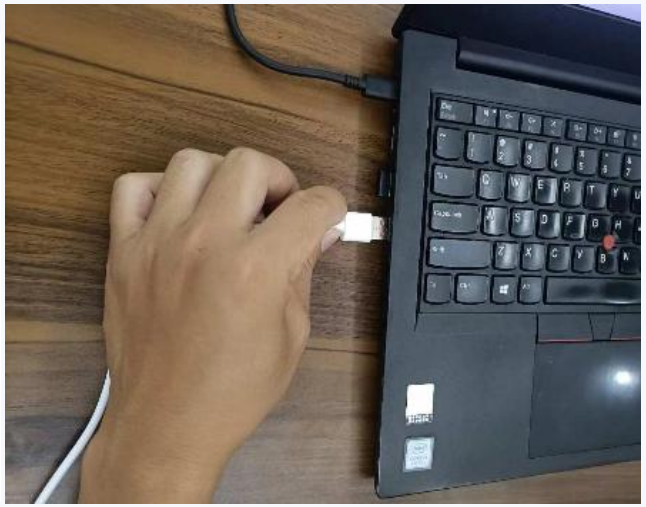
Upon acquiring raw data, the algorithm initiates a filtering process rooted in the principle of nearest similarity. Subsequently, through upsampling and Gaussian blur, the low-resolution data undergoes refinement. Finally, by identifying the largest connected regions for each anatomical part and applying area/edge-length-based criteria to eliminate artifacts, a high-precision heatmap (as depicted on the right) is generated.

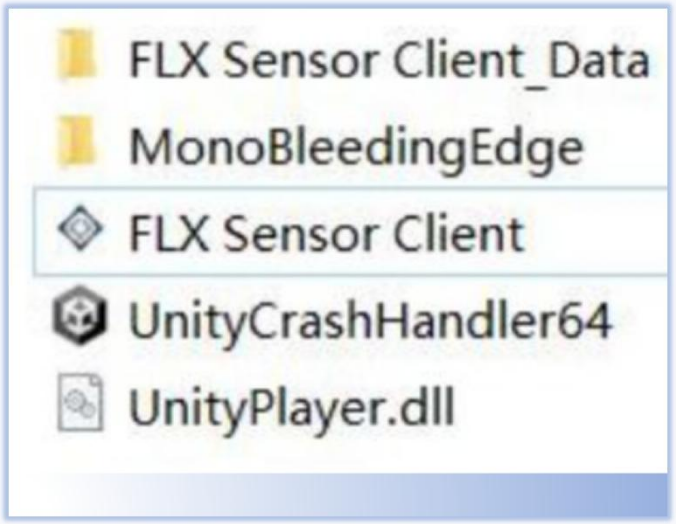


- we can provide supporting sampling hardware, upper-layer algorithms, or demonstration software If required, Stable pressure data
- graphs can be derived through the integration of high-precision heatmaps.

Step 1. How To Connect The Device Properly?

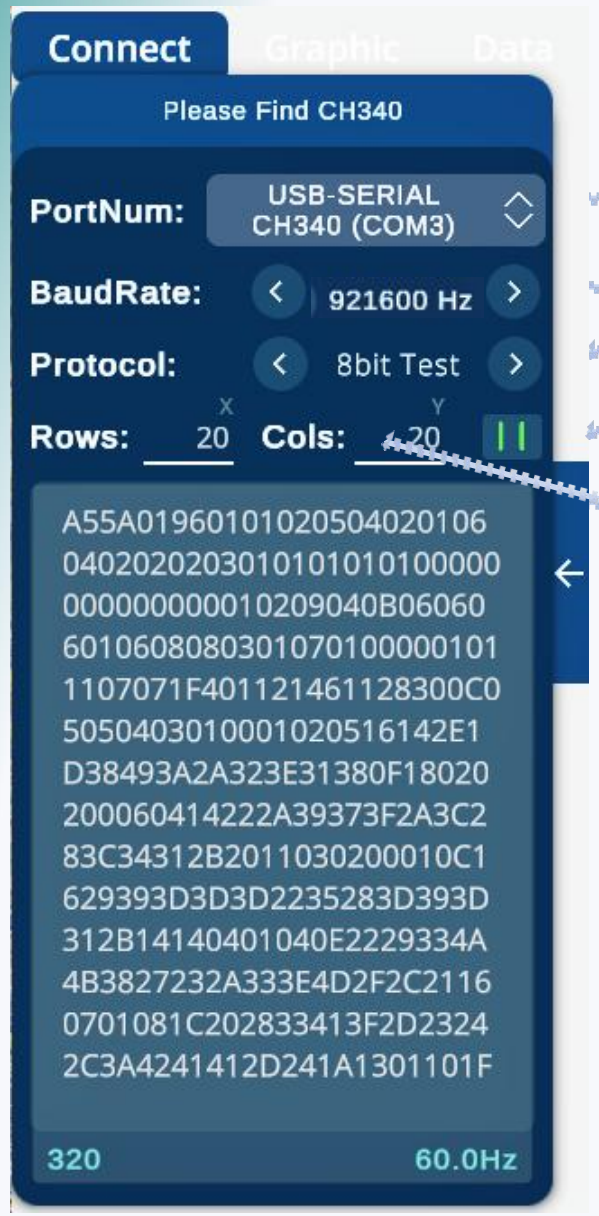






- One side connects to the Pressure Sensor Seat Cushion, and the other side connects to the computer/ laptop.
- After connecting the sensing cushion, please first click "UnityCrashHandler64" and then open "FLX Sensor Client".

Step 2. How to start the software properly?



Click here after connecting the Type- C cable Choose PortNum: USB- SERIAL CH340 (COM3)

Select baudrate: 921600Hz for Sensor Matress

Protocol: 8 bit Test

Enter the rows and columns of sensor's array: Rows: 20 Cols: 20

After setting the parameters properly, Click here, the system will start running

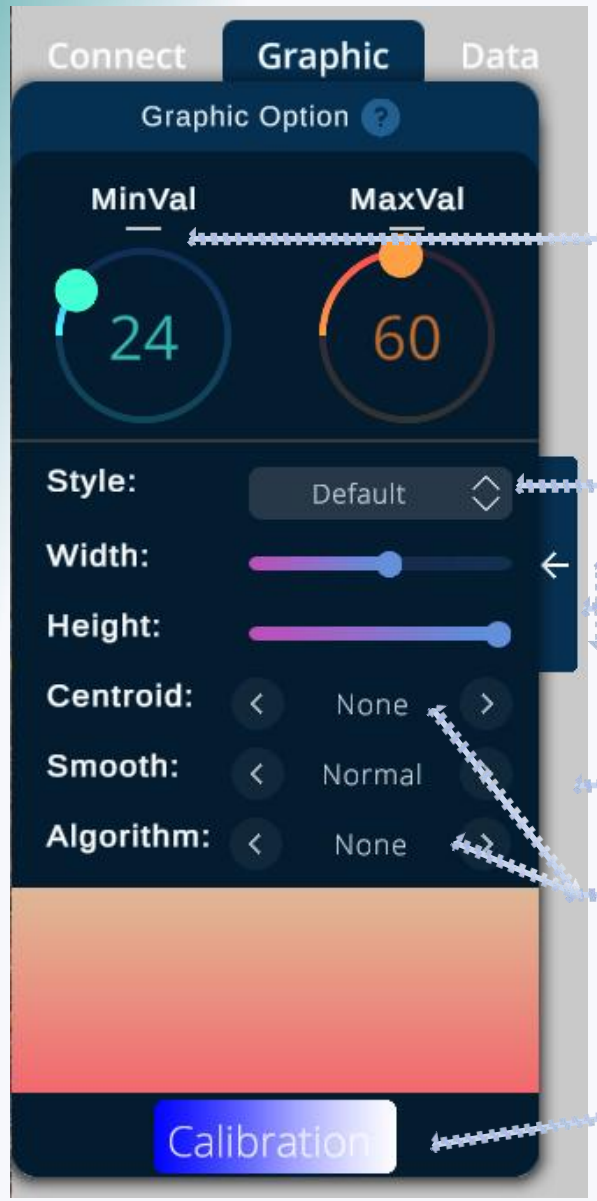
Click here to hide the interface.

Click here to Change the system language settings.

Send to Serial Port.



Step 3. Graphic settings



The image display effect can be adjusted by modifying the minval and maxval limits;

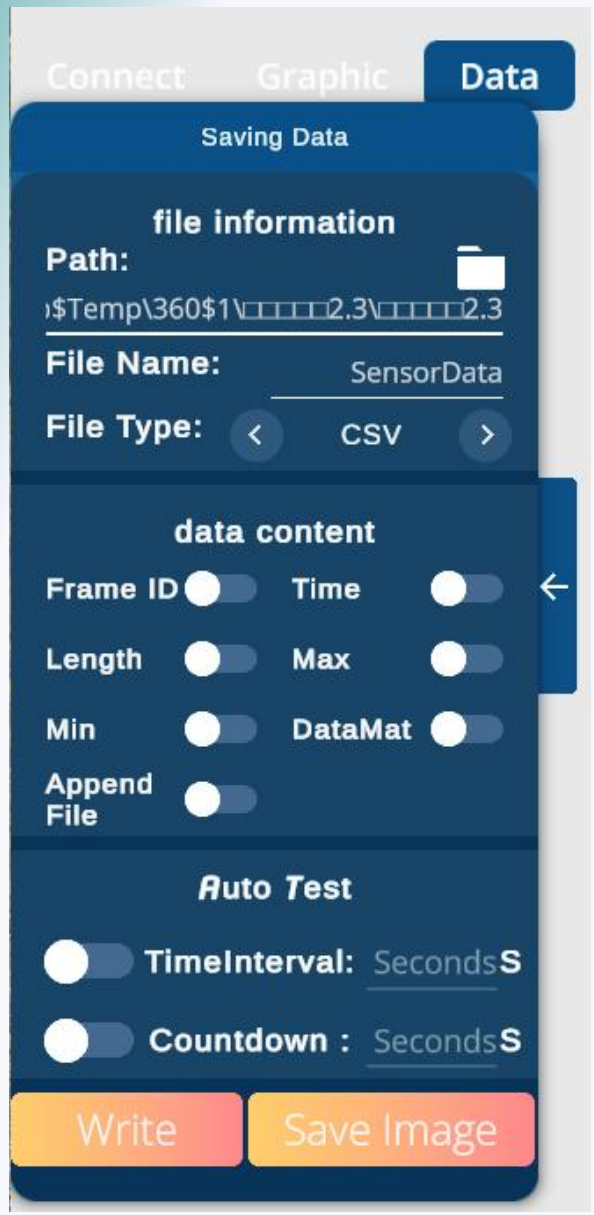
The thermal mapping diagram can be changed by adjusting the style, Pls refer to P11

The image size can be altered by modifying the Width and Height;

The edge blurring effect of the image can be controlled by adjusting the smooth. Normally set it to be “normal”.

Algorithm and Centroid and are the developer tools. Just set it to be “None”.

After completing the above operations, click “Calibration”.



Step 4. Data storage interface



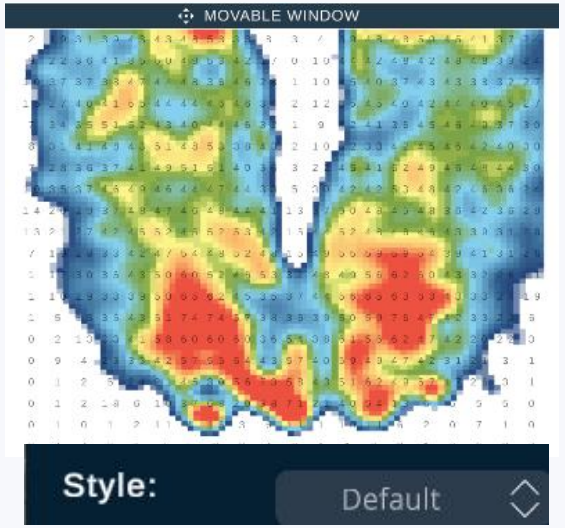
This feature is used to save real-time data. It can save data in either text format or table format. The saved values are the ADC values collected by the acquisition board(hardware) . This feature can also be understood as a developer tool.

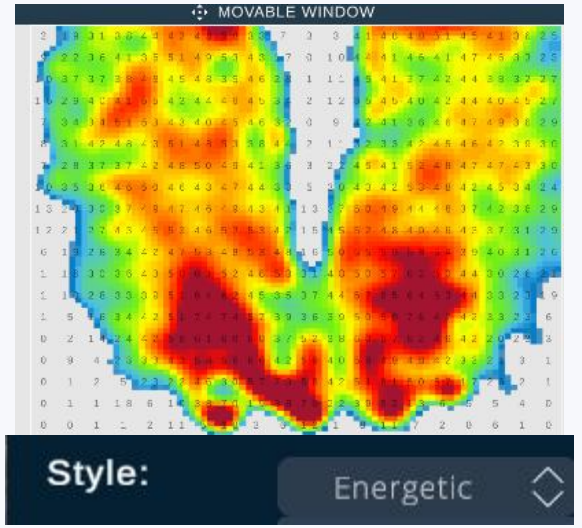


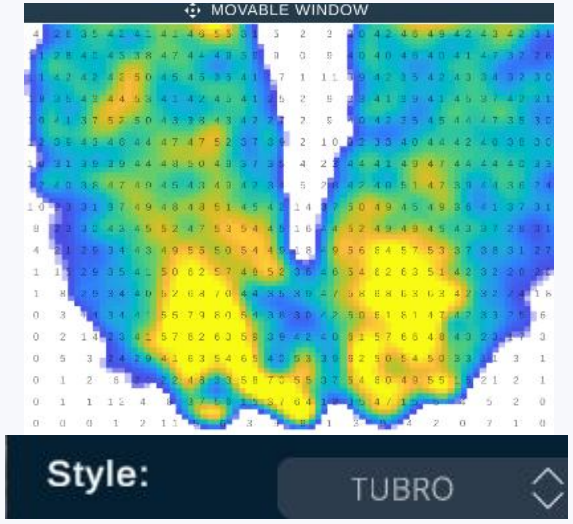
The image saving function is used to save the imagedisplayed in the image display area.

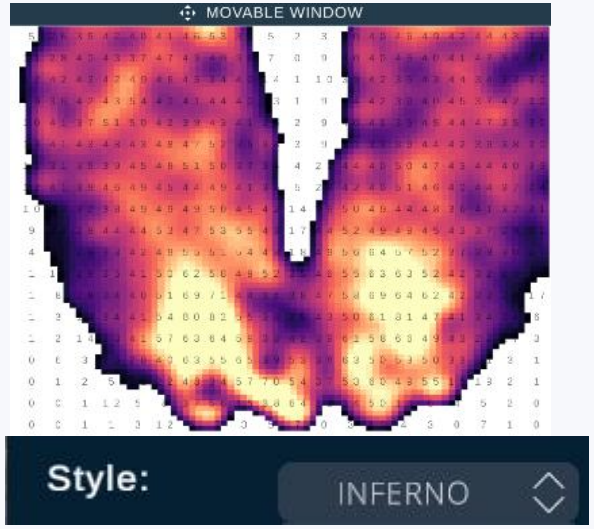
Different Mapping Style

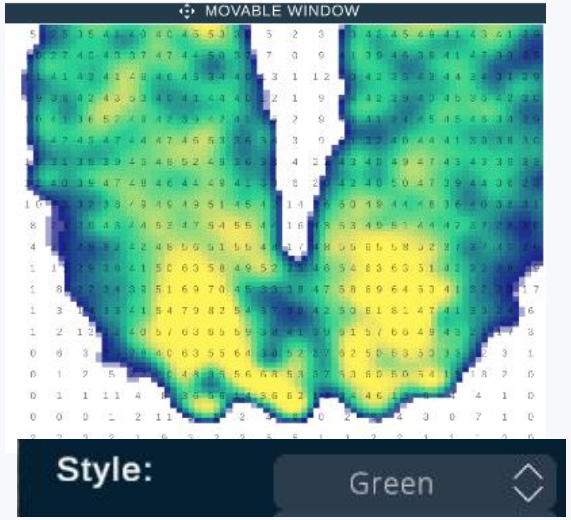
Maxrays offer five distinct thermal map modes for your selection, each featuring a unique visual style and a focused approach to data presentation. These include a high-contrast contour mode that emphasizes pressure differences, and a dynamically color-changing style that adapts to real-time data variations. This diverse range of options ensures you can select the most suitable thermal map mode based on your actual application scenarios—such as daily monitoring, professional analysis, or demonstration presentations—making pressure distribution data more intuitive, comprehensible, and tailored to your personalized needs.











Scenarios of failure ---- 1

The electrode area of the sensor is damaged

When the electrode is partially damaged, the static resistance in this area will change slightly, but it will not affect the normal operation of the sensor.

When the electrode is completely damaged, the function of this area will fail, but it will not affect the normal operation of other areas. When the electrode at the interface is damaged, all areas will fail to function.

Short circuit between positive and negative electrodes

The upper and lower electrodes of the sensor are usually composed of conductive circuits with low resistance. If the intermediate material of the sensor is damaged or displaced, it may easily cause a short circuit between the upper and lower electrodes. In this case, the resistance of the corresponding functional area will below with insignificant changes, resulting in the failure of the corresponding functional area, but it will not affect the normal operation of other areas.

Excessive deflection

In addition to pressure strain, the sensor also undergoes a certain degree of flexural strain. The internal stress generated by bending will cause a reduction in the sensor's measuring range. When the deflection reaches a sufficient magnitude, the resistance will drop to the limit value, which in turn makes the corresponding area unable to produce pressure strain.

If there is a need for installation with large deflection, prior communication is required. We can conduct targeted design on the distribution of pressure-sensitive areas—for example, avoiding locating areas with large deflection at the center of pressure-sensitive regions. Additionally, the force in large-deflection areas can be transmitted to other small-deflection areas through silicone sheets

Scenarios of failure ---- 2

High temperature

High temperatures above 50° C will cause a sharp increase in the sensor's resistance. When the sensor's resistance exceeds the reasonable range designed for the circuit, it may easily lead to sensor failure. However, the sensor will return to normal once the temperature returns to room temperature.

Irreversible thermal damage

Sustained high-temperature conditions exceeding 70° C, or localized ultra-high temperatures exceeding 100° C, may easily cause irreversible deformation of the intermediate material. As a result, the tightly bonded upper and lower electrodes may bulge or form bubbles due to the deformation of the intermediate material, which impairs the sensor performance—and such impact will persist even after the temperature returns to normal.

Currently, we offer high-temperature-resistant models for special scenarios, which can completely avoid the limitations caused by the above temperature conditions.

Fracture of the silk-screen printed circuit

Repeated bending can cause creases in the piezoresistive film, which may lead to its fracture. This may result in abnormally high resistance values in the affected area, or intermittent poor contact

Loose Terminal Connection

Frequent or forceful plugging/unplugging may compromise terminal integrity, resulting in intermittent connectivity or complete signal loss

Internal water or liquid seepage

The surface tension of liquids may, to some extent, alter the sensor’s microstructure, modify the contact resistance between the upper and lower electrodes, and result in reduced sensor sensitivity.

Frequently Asked Questions ---- 1

Is this sensor suitable for reading accurate pressure values? (Can it be used to make high-precision electronic scales?)

It is not recommended for use in weighing scenarios. Pressure sensors are generally only suitable for detecting trends in pressure changes or determining the presence or absence of pressure, rather than performing precise calculations of absolute values. Therefore, they are not suitable for application scenarios requiring high precision (such as high-precision electronic scales).

What is the accuracy of the sensor?

The specific accuracy varies based on the application scenario, as well as the contact area and material of the object. Typically, the error ranges from approximately ±10%. For higher precision, however, it is advisable to perform software calibration tailored to the actual application scenario to establish a pressure-output correspondence.

How to obtain sensor data?

The thin-film pressure sensor operates on the principle that its resistance changes under pressure. Resistance or resistance voltage division can be sampled using a chip with an ADC (Analog-to-Digital Converter). Most common consumer electronics chips come with ADC sampling capabilities.

What type of surface is suitable for attaching the sensor?

The sensor performs optimally on dry, flat, and smooth surfaces. Exposure to humid gases or dust particles may interfere with the accuracy of its data detection.

Is the sensor waterproof?

The fabric sensor itself is not waterproof, and neither is its circuit component. The outer casing of the Maxrays mattress is made of silicone cloth, which offers a certain level of water resistance and corrosion resistance.

Frequently Asked Questions ---- 2

We chat

What effect does humidity have on the sensor?

Humidity can impact the sensor if no waterproofing treatment is applied. Moist air may cause oxidation of the internal circuits, which in turn increases the resistance of the sensor's entire circuit. To counteract silver paste oxidation.

What effect does temperature change have on the sensor?

An increase in temperature causes the sensor's resistance to rise. The sensor is recommended for use in environments with temperatures between -20℃ and 50℃ That said, high temperatures up to 70℃ will not result in irreversible deformation of the sensor. Given that products are typically exposed to temperatures around 65℃ during container shipping, it can be used with confidence in common consumer-facing scenarios.

Will exceeding the sensor's measurement range damage it?

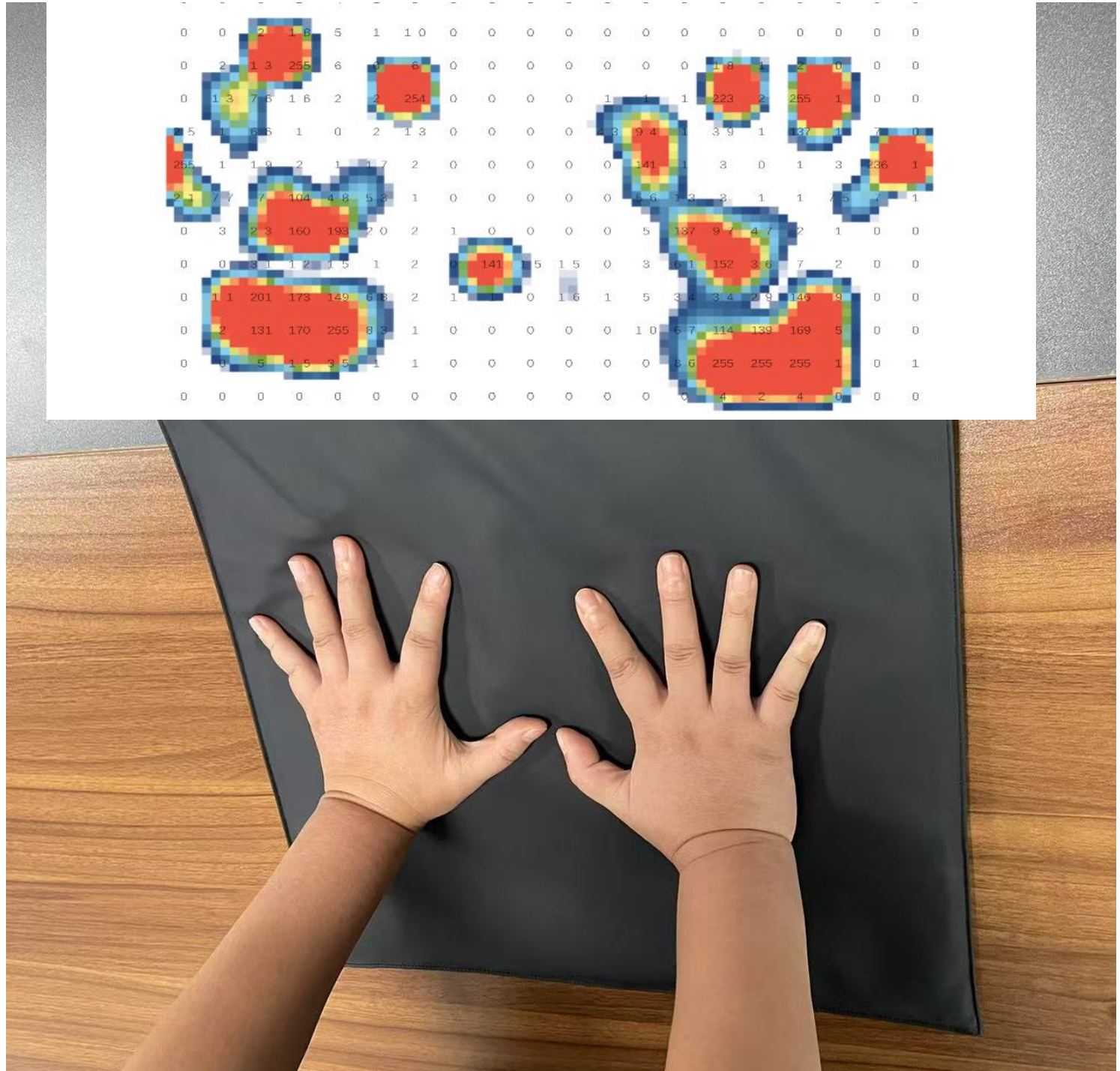
The measurement range is calibrated to the optimal linear interval. As long as no physical damage is inflicted on the sensor, exceeding this range will not compromise its testing performance—it will only affect the resolution and variation range of the current measurement.

What are the main application scenarios of the sensor?

Maxrays pressure sensor offers a cost-effective solution for a wide range of applications, including pressure and pressure intensity measurement, bending detection, touch interaction, and collision detection. Its existing solutions have been deployed in fields such as medical care, sports health, industrial control, trendy toys, and wearable devices.

Does the contact area have an impact on the sensor's readings?

While differences in contact area (e.g., minimal vs. larger) may lead to variations in sensor readings. A larger contact area contributes to more accurate results, so it is recommended that the contact area exceed 0.25 cm^2 during use. For applications involving contact with sharp objects, attaching buffer materials like foam sponge or silicone to the sensor is advised to ensure optimal testing performance.



SHENZHEN MAXRAYS TECHNOLOGY CO., LTD.

Company address: Unit A,Weidonglong Building, No.4549 Longhua Avenue Qinghua Community, Longhua Street, Longhua District, Shenzhen, Guangdong, China

Web site: maxrays.en.alibaba. com

Contact Person: Amy

Phone/Wechat: +86 157 9766 8676

Email: amy@maxrays.cn

We chat



Whats app

